

SASKATOON INTL, SK, CanadaWMO: **718660**Lat: **52.1736N**Lon: **106.7189W**Elev: **504**StdP: **95.41**Time Zone: **-6.00 (W06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-30.7	-37.9	0.1	-33.5	-34.8	0.1	-30.5	12.6	-7.5	11.2	-7.0	3.4	240	0.716

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	30.2	18.9	28.3	18.0	26.5	17.4	20.7	27.3	19.5	25.9	18.4	24.4	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	14.2	24.0	17.3	13.1	22.4	16.1	12.2	21.1	62.4	27.2	58.0	25.7	54.3	24.3	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.9	8.8	-37.4	34.0	3.3	2.4	-39.8	35.8	-41.8	37.2	-43.6	38.6	-46.0	40.4		
(5)			-37.4	23.2	3.0	1.3	-39.5	24.1	-41.3	24.8	-43.0	25.5	-45.2	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-15.1	-13.5	-6.2	3.9	10.9	15.6	18.4	17.4	12.2	4.1	-5.2	-12.7
	DBStd	13.54	8.78	7.82	7.97	5.84	4.67	3.60	3.14	3.66	4.67	5.07	6.89	8.05
	HDD10.0	3563	779	657	502	193	46	3	0	1	29	193	456	705
	HDD18.3	5853	1037	891	761	432	234	96	38	63	191	443	705	963
	CDD10.0	849	0	0	0	11	73	172	262	228	94	9	0	0
	CDD18.3	97	0	0	0	0	3	15	41	33	6	0	0	0
	CDH23.3	1333	0	0	0	6	83	208	467	439	125	3	0	0
	CDH26.7	368	0	0	0	1	17	51	127	139	33	0	0	0
Wind	WSAvg	4.3	4.2	4.2	4.5	4.9	4.9	4.4	3.9	3.9	4.2	4.3	4.2	4.0
Precipitation	PrecAvg	364	17	14	17	21	46	66	58	39	33	18	15	19
	PrecMax	486	43	32	48	62	150	162	147	91	70	76	40	34
	PrecMin	236	5	1	4	2	8	12	14	3	5	1	1	6
	PrecStd	58	9	9	11	15	32	36	33	24	18	17	10	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	3.8	12.7	23.0	28.8	31.1	32.5	33.4	30.3	22.5	12.8	5.2
		MCWB	1.7	1.7	6.7	12.1	15.5	17.6	20.9	19.0	17.5	13.2	7.6	2.5
	2%	DB	2.1	1.5	8.6	19.8	25.7	27.9	29.8	30.1	26.9	18.1	8.3	2.7
		MCWB	0.5	0.1	4.8	10.1	14.1	17.0	20.0	18.8	16.1	10.6	4.8	0.8
	5%	DB	-0.1	-0.2	5.6	16.8	23.0	25.7	27.8	27.8	24.0	15.0	5.4	0.4
		MCWB	-1.3	-1.2	3.0	8.9	12.9	16.3	19.1	18.2	14.8	9.0	2.9	-0.9
	10%	DB	-3.0	-2.3	3.3	13.8	20.6	23.6	26.0	25.7	21.2	12.1	3.4	-2.1
		MCWB	-3.9	-3.1	1.4	7.3	11.7	15.4	18.4	17.3	13.4	7.5	1.5	-3.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	2.0	7.5	12.7	17.0	20.9	23.3	21.4	18.6	13.7	7.9	2.7
		MCDB	3.6	3.4	11.8	21.2	24.5	27.0	30.0	28.5	27.3	21.5	12.2	4.9
	2%	WB	0.6	0.3	5.0	10.9	15.0	18.7	21.4	20.1	16.8	11.3	5.1	1.0
		MCDB	2.0	1.3	8.4	18.0	22.9	24.8	27.5	27.8	25.0	16.4	8.0	2.7
	5%	WB	-1.3	-1.2	3.1	9.4	13.6	17.4	20.1	18.9	15.4	9.5	3.1	-0.9
		MCDB	-0.1	-0.2	5.5	16.1	21.7	23.4	26.1	25.7	22.5	14.1	5.0	0.4
	10%	WB	-3.9	-3.1	1.5	7.7	12.3	16.3	19.0	17.9	14.0	7.9	1.6	-3.0
		MCDB	-3.1	-2.3	3.1	13.1	19.4	21.9	24.6	24.3	19.9	11.5	3.3	-2.1
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.1	9.9	12.0	13.7	12.0	12.5	13.5	13.3	11.3	9.0	8.9
		MCDBR	10.7	10.9	11.5	16.9	17.5	15.6	15.4	17.0	18.5	16.2	11.6	10.5
		MCWBR	9.4	9.6	8.0	9.3	8.5	7.1	7.3	7.4	9.0	9.6	8.3	9.0
	5% WB	MCDBR	10.6	10.5	11.1	15.9	15.5	12.9	13.5	14.6	16.6	15.0	10.9	10.5
		MCWBR	9.4	9.4	7.9	9.1	7.9	6.7	7.1	7.0	8.5	9.1	8.0	9.0
Clear-Sky Solar Irradiance	taub	0.251	0.266	0.296	0.343	0.357	0.377	0.377	0.379	0.324	0.282	0.253	0.219	
	taud	2.285	2.275	2.349	2.339	2.346	2.338	2.342	2.330	2.469	2.560	2.473	2.375	
	Ebn at Noon	766	861	903	893	894	875	869	845	863	848	779	770	
	Edh at Noon	62	84	97	112	117	119	116	112	86	64	51	46	
All-Sky Solar Radiation	RadAvg	1.11	2.15	3.63	4.79	5.89	5.96	6.27	5.09	3.69	2.17	1.09	0.80	
	RadStd	0.10	0.13	0.27	0.47	0.46	0.47	0.35	0.38	0.35	0.32	0.13	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page